Barnita Sharma

ee19resch11002@iith.ac.in | barnitasharma@gmail.com | +91 8638385360 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	8.17
Swadeshi Academy, Guwahati	Class XII (AHSEC)	2010	73.20%
Disneyland High School, Guwahati	Class X (SEBA)	2008	85.00%

SKILLS

Technical Skills:

- **Domain Expertise:** 3D Computer vision, Image processing, Machine learning, Deep learning
- **Programming Languages:** Python, C++, C
- **Tools & Frameworks :** Python Libraries: [PyTorch, PyTorch3D, Open3d, OpenCV, NumPy], 3D modeling and visualization tools: [Meshlab, Blender]

Communication Skills:

- **Technical Writing:** Proficient in creating comprehensive documentation, research papers, and project reports
- **Oral Communication:** Experienced in delivering presentations, seminars, and lectures to diverse audiences
- **Visual Presentations:** Skilled in developing engaging slide decks to enhance audience engagement

POSITIONS OF RESPONSIBILITY

Junior Research Fellow (July 2018 - April 2019)

Indian Institute of Information Technology, Guwahati

Project: *Development of a Low-Cost, Portable Telepathology System for Smart Villages*

Responsibilities:

- **System design:** Developed a portable telepathology system by integrating a Foldscope, mobile phone camera, and Raspberry Pi to to capture and transmit high- resolution histopathological images of breast tumor biopsy samples. Employed computer vision techniques, including Gabor filters for feature extraction and an SVM- based classifier, to differentiate between benign and malignant samples as part of an initial diagnostic process.
- **Workshops and Publication:** Delivered workshops on Foldscope usage to healthcare workers and students. Co-authored a research paper published in an IEEE conference detailing the system's development and its application in rural healthcare.

RELEVANT COURSES

- Foundations of machine learning
- Detection and Estimation theory
- Random variables and Random process
- Introduction to modern AI

PROJECTS

Ph.D. Thesis (In Progress)

Title: *Image-Based 3D Reconstruction and Rendering of Cultural Heritage Sites*

Overview: Developing methods to create geometrically accurate and visually realistic 3D models of cultural heritage sites from unstructured internet imagery, aimed at supporting digital preservation and immersive virtual tourism experiences.

Contributions:

- Developed efficient 3D reconstruction methods that produce accurate and smooth models
- 2D part segmentation combining sparse 3D reconstruction with segmentation models to obtain consistent part labels across diverse views.
- 3D part reconstruction that lifts 2D part labels into complete 3D volumes using symmetry and semantic consistency.
- Created immersive, high-quality visualizations of reconstructed monuments for digital heritage applications:
- Developed a hybrid 3D-2D visualization pipeline leveraging feature matching and pose alignment to recover fine architectural details beyond geometric reconstructions.
- Integrated a VR add-on and web-based deployment to enable immersive, cross-device access to heritage visualizations

M. Tech. Dissertation

Title: *Facial Expression Recognition using Convolutional Neural Networks*

Overview: Developed a system to automatically recognize human facial expressions using deep learning techniques, with applications in human-computer interaction and psychological analysis.

Methodology:

- Utilized the FER-2013 dataset containing images across seven emotion categories (Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral), and applied the Viola-Jones face detector to isolate facial regions and standardize input, reducing background noise.
- Designed and implemented a custom three-layer Convolutional Neural Network (CNN) with a softmax output to classify expressions based on raw, extracted facial regions.
- Results showed that happiness was more accurately classified due to its high dissimilarity from other expressions, while subtle emotions like fear and surprise were often confused, highlighting the difficulty in distinguishing between similar expressions.

B.E. Final Year Project

Title: *Detection of Venation in Leaves using various Edge Detection Algorithms*

Overview: Developed a system to detect and analyze venation patterns in leaves, contributing to plant phenotyping and botanical studies.

Methodology:

- Implemented and compared three well- known edge detection algorithms (Prewitt, Sobel, and Canny) to extract vein patterns from a custom image dataset, evaluating the strengths and limitations of each approach.
- Developed a MATLAB-based GUI to automate the process, allowing users to select images and obtain detected vein patterns with minimal input.

PUBLICATIONS

1. B. Sharma, S. Jana, *et al.*, "Efficient 3D Reconstruction of Heritage Monuments Using Symmetry-Aware Part-Based Modeling," *The Visual Computer*, Accepted for publication (2026). Preprint available: Research Square, Dec. 2025. DOI: https://doi.org/10.21203/rs.3.rs-8416098/v1
2. B. Sharma and S. Jana, "Semantic Part Segmentation of Heritage Monuments from Crowdsourced Images: Consistent Labeling using SfM-guided SAM," *2025 National Conference on Communications (NCC)*, 2025, pp. 1-6. DOI: https://doi.org/10.1109/NCC63735.2025.10983255
3. B. Sharma, J. Sai Sri Hari Vamshi, S. Jajula, H. K. Patel, and S. Jana, "VR Visualization of Heritage Structures at Different Scales via Mode Switching," *2023 IEEE 20th India Council International Conference (INDICON)*, Hyderabad, India, 2023, pp. 349-354. DOI: https://doi.org/10.1109/INDICON59947.2023.10440757
4. R. Patra and B. Sharma, "Development of a Low-Cost, Portable Telepathology System for Smart Village using Foldscope," *2022 IEEE International Symposium on Smart Electronic Systems (iSES)*, Warangal, India, 2022, pp. 574-577. DOI: https://doi.org/10.1109/iSES54909.2022.00125

PROFESSIONAL SUMMARY

Ph.D. candidate specializing in Computer Vision, with research experience in combining classical geometry and deep learning for image-based 3D reconstruction, semantic segmentation, and visualization of cultural heritage sites. Experienced in developing computationally efficient research pipelines for 3D modeling and immersive visualization from real-world imagery. Seeking industry roles in computer vision, 3D perception, or AR/VR technologies.